

1. "How Do Big Systems Store Data?":

== instagram: primarily utilizes PostgreSQL, a relational SQL database, for structured data storage.

##@#### ----For example :

USER	
Pk	ID
	First_name
	second_name
	third_name

Post	
PK	ID
FK	created_by_user_id

Uber employs a combination of SQL and NoSQL databases to cater to different data needs.

Amazon developed **DynamoDB**, a NoSQL database, to handle massive scalability and performance requirements.[Wikipedia+1PREP INSTA+1](#)

#--Database Structures Used:

- **SQL (Relational Databases):** Ideal for structured data with complex relationships. They offer ACID compliance, ensuring data integrity.
- **NoSQL (Non-Relational Databases):** Suitable for unstructured or semi-structured data, providing flexibility and horizontal scalability.[GeeksforGeeks](#)

#--Example Using SQL Server:

While many large tech companies have shifted towards open-source or proprietary solutions, **Microsoft SQL Server** is still prevalent in enterprises, especially those integrated into the Microsoft ecosystem.

Task 2: Do Video Games Use Databases?

Research Question: How do online games (like PUBG, Fortnite, etc.) store player progress, points, and profiles?

Real-World Application: Online games require efficient data storage for player profiles, game progress, and in-game assets.

- **Data Storage Methods:**

- **Relational Databases (SQL):** Used for structured data like user accounts and transaction histories.(enitity..) (weak entity or strong entity)
- **NoSQL Databases:** Employed for unstructured data, such as game states and real-time analytics.

Example of a Game Using Relational Database:

Many games utilize **MySQL** or **PostgreSQL** for managing user data due to their robustness and reliability.

Task 3: Can a Database Work Without the Internet?

[#--Research Question: How are databases used in offline systems, like in military bases or airplanes?](#)

Real-World Application: In environments where internet connectivity is unavailable or restricted:

- **Offline Databases:** Systems like **Microsoft SQL Server** can operate entirely offline, storing data locally and synchronizing with central servers when connectivity is restored.

Use Cases:

- **Military Bases:** Secure data storage without external connectivity.
- **Airplanes:** In-flight entertainment systems and operational data logging

Task 4: What Makes Online Stores So Fast?

Research Question: How do e-commerce platforms (like Amazon or Noon) organize their product and order data for fast searching?

Real-World Application: E-commerce platforms manage vast product catalogs and high user traffic.

- **Data Organization:**

- **NoSQL Databases:** Used for product catalogs to allow quick searches and scalability.
- **Relational Databases:** Employed for transactional data to ensure consistency.
- **Cloud Infrastructure:** Platforms like Amazon Web Services (AWS) provide scalable and distributed systems.

Task 5: How Do Banks Protect Their Data?

Research Question: How do banking systems use databases to protect sensitive customer data?

Real-World Application: Banks handle confidential financial data requiring stringent security measures.

- **Database Features for Security:**

- **Encryption:** Protects data at rest and in transit.
- **Access Controls:** Ensures only authorized personnel can access sensitive data
//not like file based they won't be able to do this ..
- **Audit Trails:** Tracks data access and modifications.

Example Using Oracle or SQL Server:

Many banks utilize **Oracle Database** for its advanced security features and **SQL Server** for seamless integration with Microsoft services.

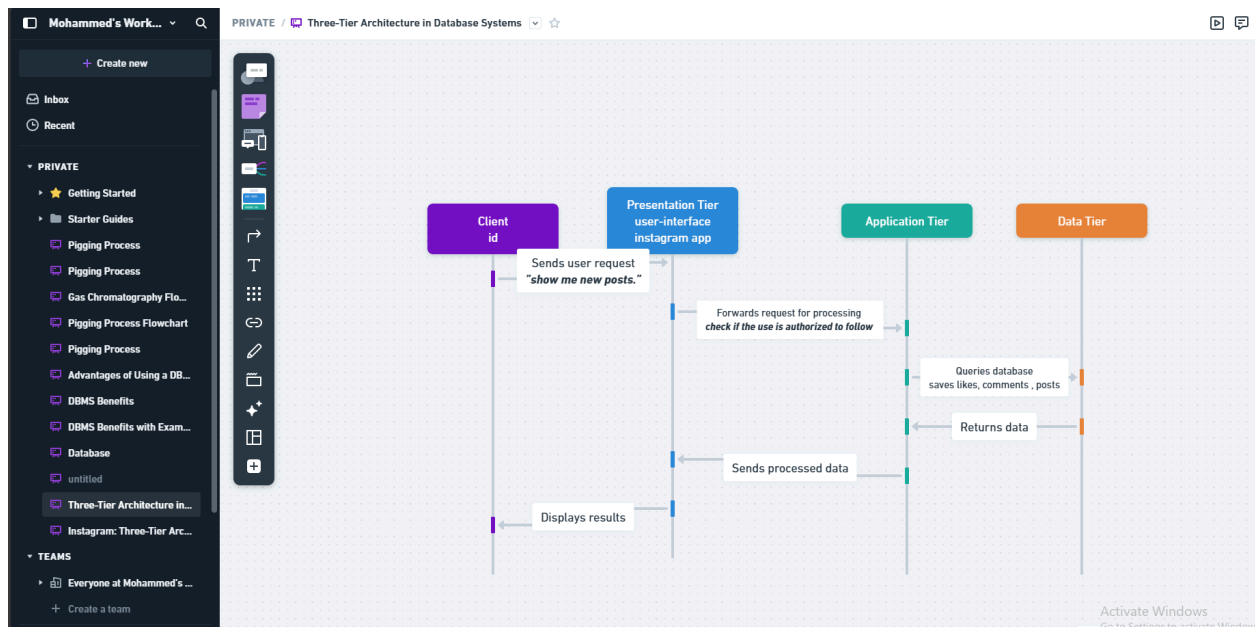
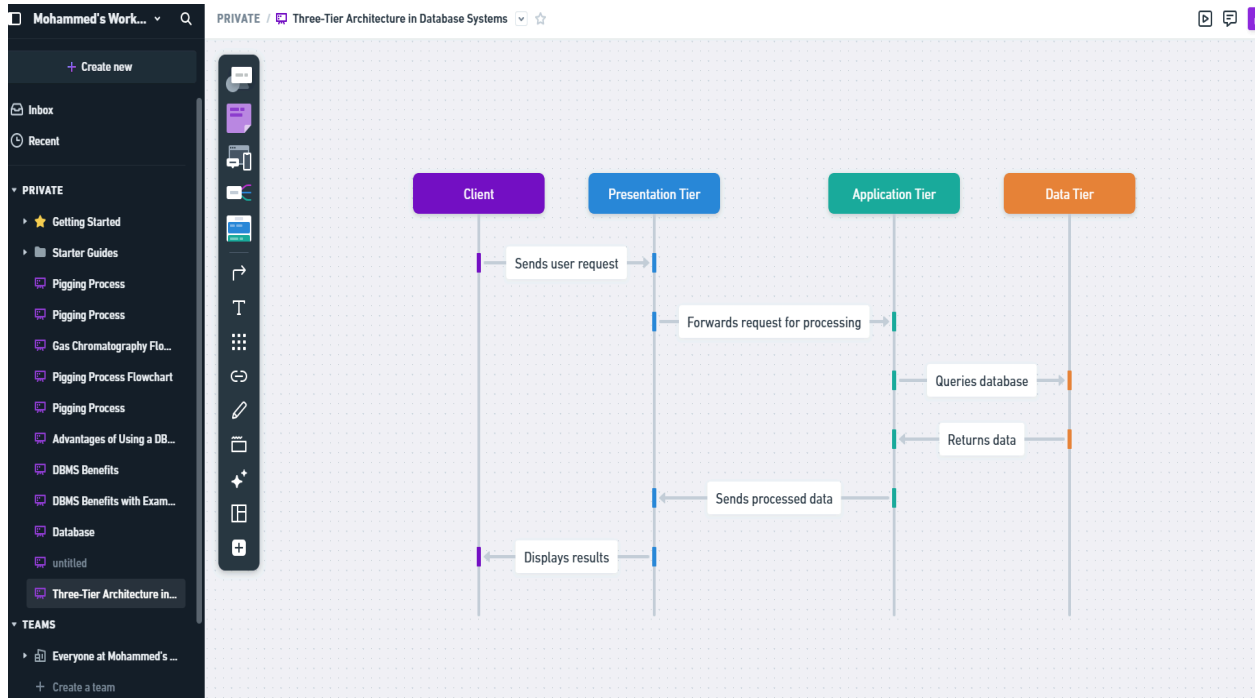
Task 6: Draw Me the System!

Research Question: What is the three-tier architecture in database systems?

Real-World Application: Three-tier architecture separates concerns, enhancing scalability and manageability.

Report (DataBase)

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#--Reference :

<https://whimsical.com/three-tier-architecture-in-database-systems-KYooFY9GdT4zbpcZMbN7rm>

